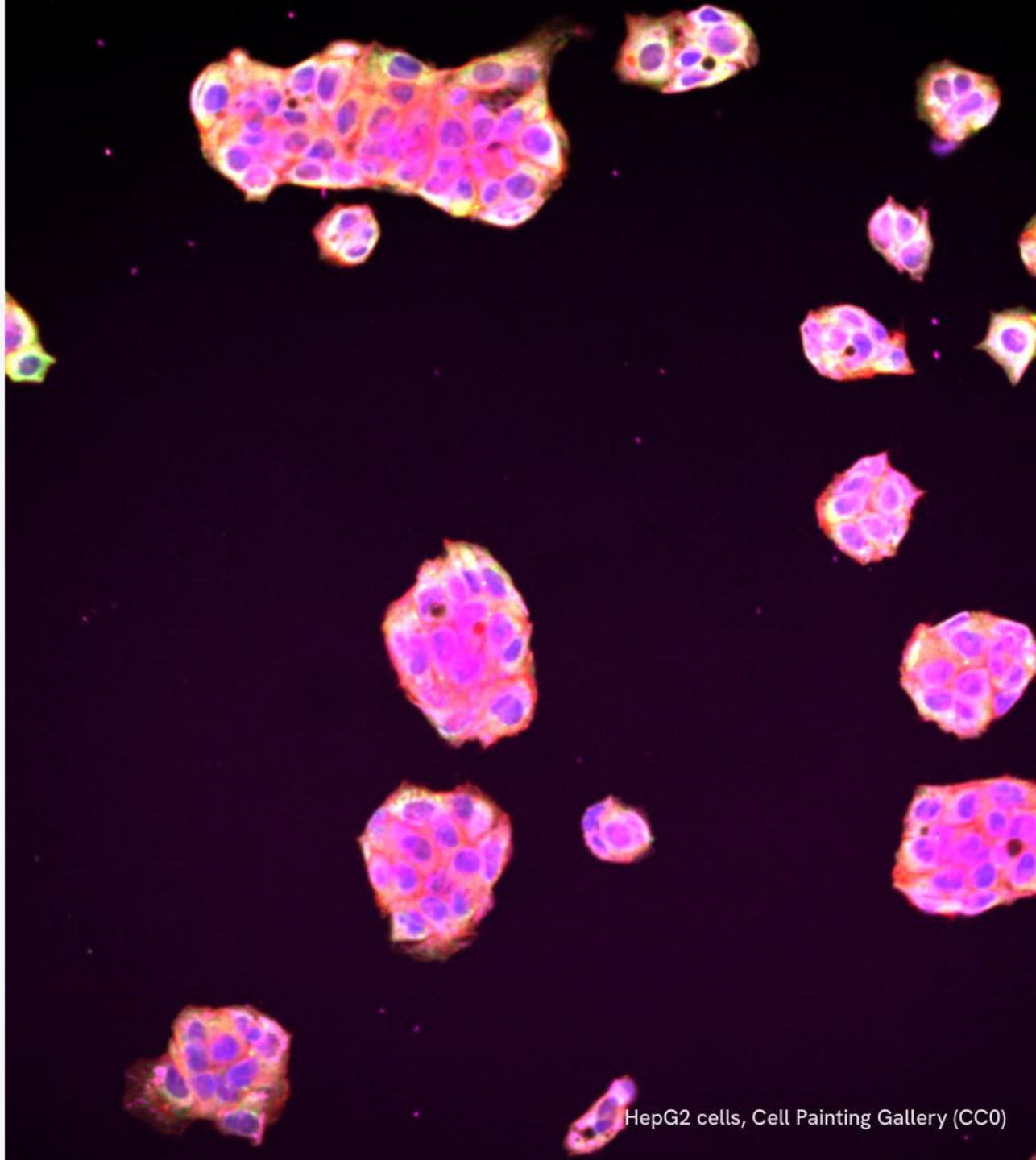
AI for Life Science Challenge · Finals

Beyond the Count

A calibrated audit for AI toxicity models built on Cell Painting

Team BioBytes

Vrushti · Dhairya



HepG2 cells, Cell Painting Gallery (CC0)

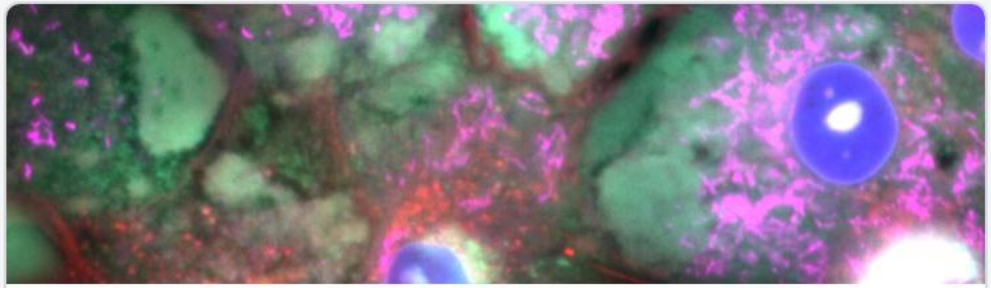
A toxicity model can look smart while only counting cells

31 *of* 49

endpoints well predicted by gene expression were predicted equally well by cell count alone

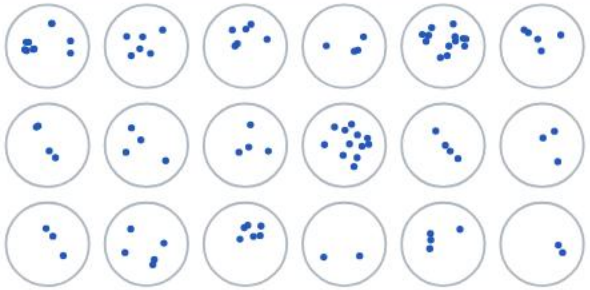
Seal et al., *Nature Communications*, 2026

- Cell Painting: six dyes, five channels, thousands of features per cell
- Many toxicity labels track cytotoxicity, so counting cells can score well
- Cell-count-only AUC 0.68 against 0.67 for a full-profile network
- FDA's 2026 draft guidance asks for technical characterization



What we hope the model reads

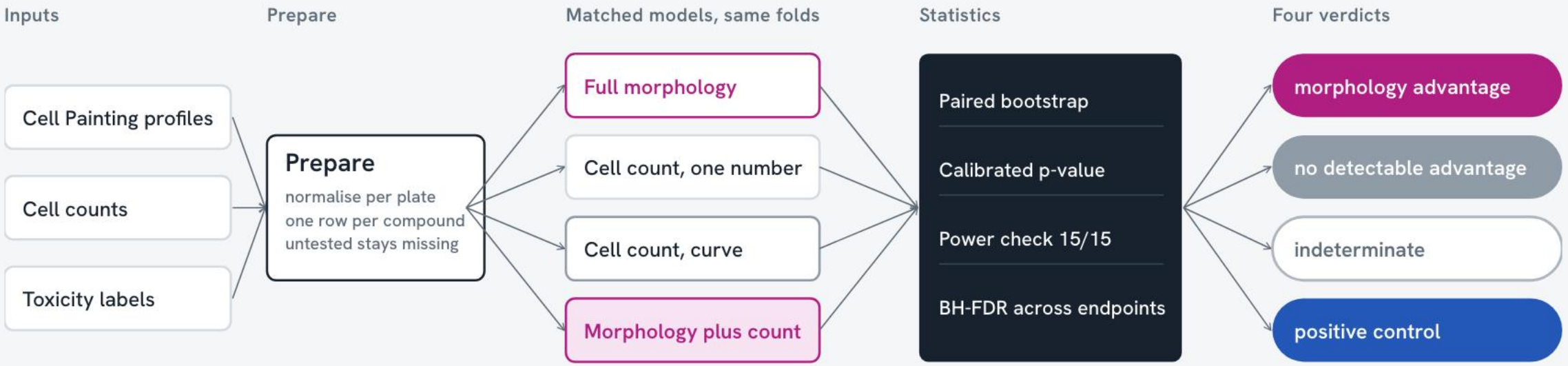
shape, texture, organelles



What it may be reading

how many cells are left

One audit that tells you, endpoint by endpoint, whether morphology adds anything



A strong baseline

Cell count sees the whole dose-response curve

Calibrated statistics

p-values checked against a label-permutation null

Power-aware verdicts

Too little data is indeterminate, never "no effect"

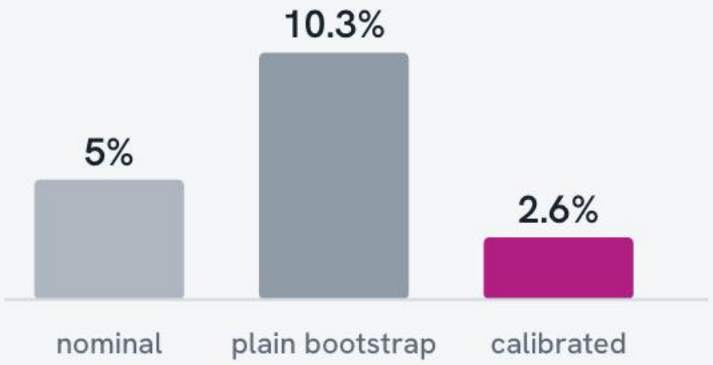
Label provenance

An untested pair is missing, never a negative

We tried to break the audit before asking anyone to trust it

Calibrated, not assumed

- Shuffle the labels, rerun everything
- Plain bootstrap: 10.3% false positives
- Calibrated: 2.6% on held-out runs



Shuffled endpoints wrongly certified, nominal 5%. CP-CNN 6.5%, nested test 6.2%.

Powered, or silent

- 15 actives and 15 non-hits to get a verdict
- Fewer means indeterminate: 223 of 405 endpoints
- Benjamini-Hochberg FDR across all powered endpoints

Clean labels

- Untested is missing, never negative
- 107,103 of 281,196 label cells observed: 62% missing
- Filtered binary labels only, found by reading the authors' notebook

Fixed seeds · MD5-verified downloads · 69 tests · runs on a laptop CPU

Morphology earns its cost on 15 endpoints, and the audit shows which



15 morphology advantage **166** no detectable advantage **223** indeterminate

1 control

- MT certified with all three representations
- Nested test certifies 23, all 15 included

405

endpoints tested

181

enough data to judge

15

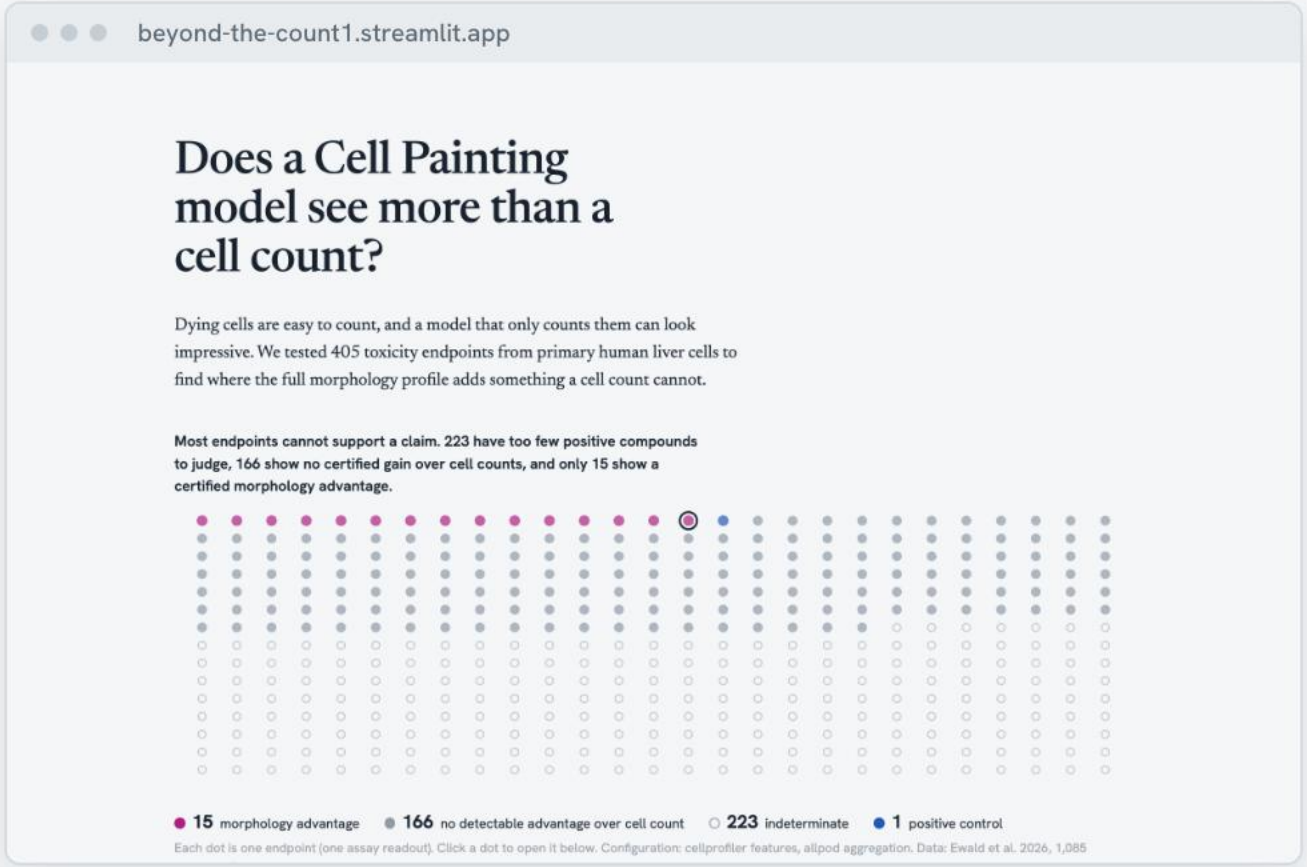
certified morphology advantage

8

certified under all three representations

A plain bootstrap would have claimed 64. A second source (HepG2, exploratory) agrees on MT.

A working tool anyone can run, not a one-off analysis




```python
from cpsa.api import shortcut_audit
res = shortcut_audit(profile, baseline,
labels)
res.table   # one verdict per endpoint
```

- Three inputs: profiles, cell counts, labels
- One call, one verdict per endpoint
- CPU only, open licences, no paid APIs

Live demo walk-through

Verdict table | Enrichment | Chip planner

github.com/dhairya-pandya/beyond-the-count

It changes a decision for each team that touches an imaging-based toxicity readout

Phenotypic AI team

Reports accuracy and moves on

Go/no-go check before accuracy is shown

Screening lead

Which endpoints justify full profiling?

Verdict table: count readout or morphology

Organ-on-chip group

How many compounds does a study need?

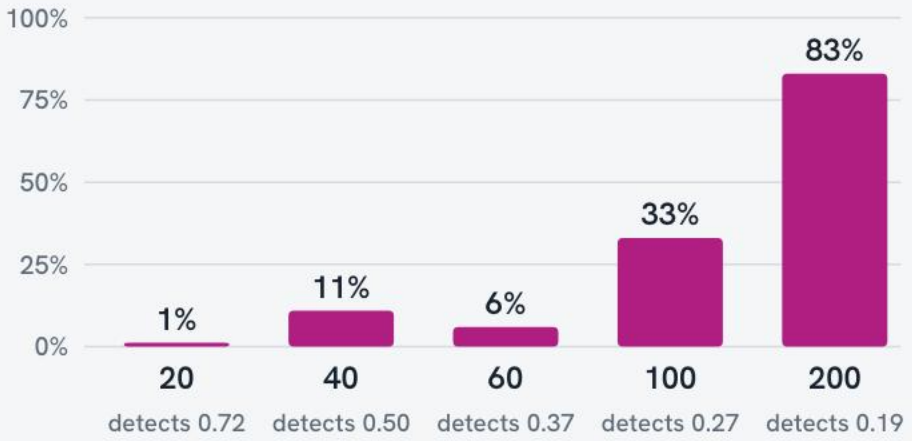
Planner sizes the study before the chip runs

Assay validation, regulatory

What supports a non-animal method?

A repeatable technical-characterization artifact

At chip scale (simulation)



Power to certify a true advantage, by number of compounds tested (20 to 200). False credit stays at 0 to 2%. No chip data used.

What we proved

2.6%

False certification, calibrated

Shuffled labels, nominal 5%. The plain bootstrap gives 10.3%

3 of 3

MT certified in every representation

CellProfiler, CP-CNN and DINOv2, with q of 0.001 to 0.039

23

Endpoints beyond cell count

Nested test certifies 23, and all 15 of the main audit are among them

0.51

Layout and batch carry no signal

Plate, well and batch alone score at chance; grouping by compound keeps replicate chips honest

The path from a working audit to something labs rely on

Next

Close the evidence gaps

- More cell types, donors, time points
- Organ-on-chip data from a partner

Gate
Verdict holds on data we did not build

→

Then

Put it in other hands

- Pilot with a Cell Painting screening group
- Their data, their labels

Gate
Outside team gets a verdict unaided

→

Later

Make it routine

- Packaged release, docs and tests
- Validation-report export
- Plate, well and batch checks

Gate
Standard pre-flight step

We would value introductions to groups running Cell Painting or organ-on-chip screens.

Close

Before you trust a phenotypic AI readout, check what it is reading

Live demo
beyond-the-count1.streamlit.app

Code
github.com/dhairya-pandya/beyond-the-count

Data
Zenodo 10.5281/zenodo.17067683

Team BioBytes. We would like to hear where it breaks.



Robustness checks, all consistent with the main result

- Power threshold 10 / 15 / 20 / 30: 19 / **15** / 22 / 26 certified
- Aggregation all / allpod / allpodcc: 18 / **15** / 3
- Chemical-similarity folds (921 clusters): all 15 stay certified
- Plate, well and batch alone: AUROC 0.51, chance
- Null width 1.0 to 1.65: 67 down to 5 certified
- Paper's own protocol: AUROC within +0.004 and +0.007
- Simulation with known truth: false credit 2 to 4%, true effects found
- Logistic regression comparator: median AUROC 0.62 against 0.65

MT, LDH and the second source

The paper's own readouts

- MT: strong count 0.873, full 0.916, q 0.019 / 0.039 / 0.001
- LDH: strong count 0.957, full 0.969, q 0.27 / 0.29 / 0.076
- Against the one-number baseline LDH looks like a win; against the curve it is not
- 38 of 51 wins vanish against the strong baseline

HepG2 against primary hepatocytes

- 327 shared compounds, 49 endpoints, one cell-count number
- Certified: 16 with HepG2, 1 with primary
- MT and LDH agree; 10 of 16 also certified in the main audit
- Weaker HepG2 baseline (one 10 µM count), so counts are not comparable

The audit at chip scale, simulated

Compounds	Cannot be judged	False credit, raw / calibrated	Power	Smallest detectable advantage
20	100%	0% / 0%	1%	0.72
40	54%	2% / 0.5%	11%	0.50
60	29%	4% / 0.5%	6%	0.37
100	1%	8% / 1.5%	33%	0.27
200	0%	8% / 2%	83%	0.19

- Replicate chips: random labels reach AUROC 0.85 if each chip is independent, 0.51 if grouped by compound
- A chip changes labels, the 3D count proxy, features, grouping and sample size